



Analysis of Water Quality Using Light Absorption Spectroscopy

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Abstract

The ocean biomes provide for both aquatic and human life systems. Water quality within the near-shore ecosystems is paramount to the perpetuation of these interdependent systems. The ability to verify water quality is the first step to understanding the interactions of various contaminants within these ecosystems. This study focuses on the applications of light absorption spectroscopy and the Beer's Law to identify possible contaminants and their concentrations within ocean water samples. The use of a Genesys 10S UV/VIS spectroscope, xenon flash bulb, and quartz cuvettes were used to identify absorptions spectra of possible contaminants. To identify the molecular profile of the samples, the method of comparing the sample absorption spectra and absorption coefficient with a control spectra and the found spectra of various possible contaminants was utilized. To determine the concentration of the possible contaminants within solution, the molar absorptivity coefficient was determined from the spectra of the various concentrations of contaminants in artificial salt water using partial least squares analysis of absorption vs concentration.

Introduction

Spectroscopic analysis of ocean water samples has been an ever increasing field of inquiry since spectroscopy has demonstrated an increased accuracy in parameter detection and reduction in time and resources used for sample analysis. The use of Principal Component Analysis and Multilinear Regression Analysis to define the connection between the ultraviolet and visual spectra and salinity values up to 50 parts per mille (ppt) has been demonstrated in controlled laboratory settings by Mecozzi and Visco^[1]. A multiparametric analysis including presence of nitrate, COD, and total suspended solids (TSS) of artificial sea water samples has been demonstrated by Hu, Wen, and Wang^[2].

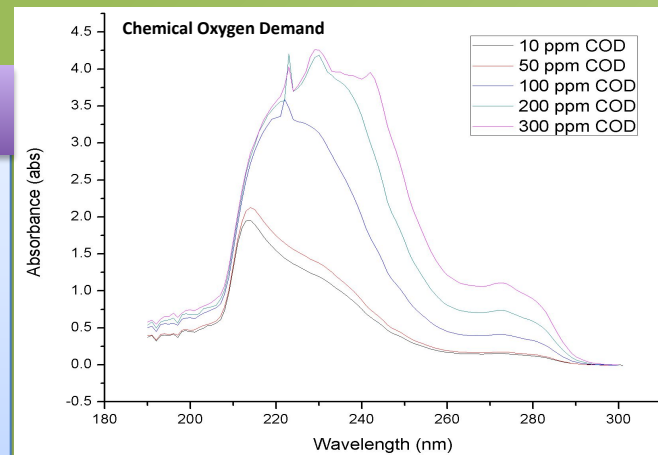
These studies provided proof of concept and built the foundation upon which this current research was carried out. One aim of this paper is to address the findings of the previous studies with dedicated highly sensitive instrumentation and apply the principles to analysis of samples taken from various near-shore ecosystems on the island of Maui. Nitrates were determined to have characteristic changes in absorption in the 215-225 nm range, and COD concentration analysis showed characteristic peaks in the 214-229 nm and 270-275 nm ranges. The analysis of taken oceanwater samples provided absorption peaks within the established wavelengths of the contaminants pointing to the presence of Nitrates and COD within the samples. Further analysis using alternative methods is being undertaken to

References

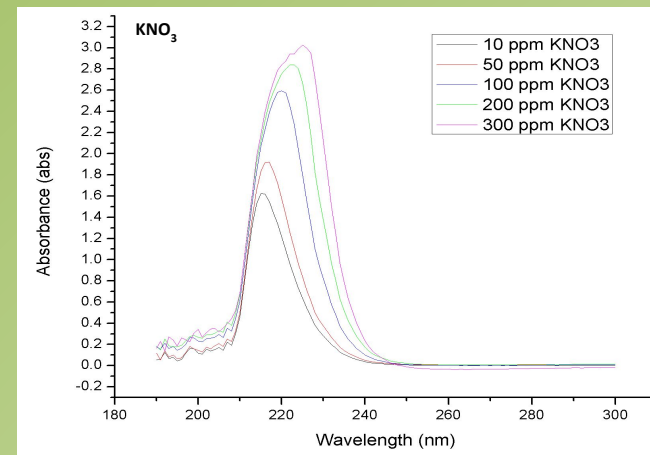
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Acknowledgments

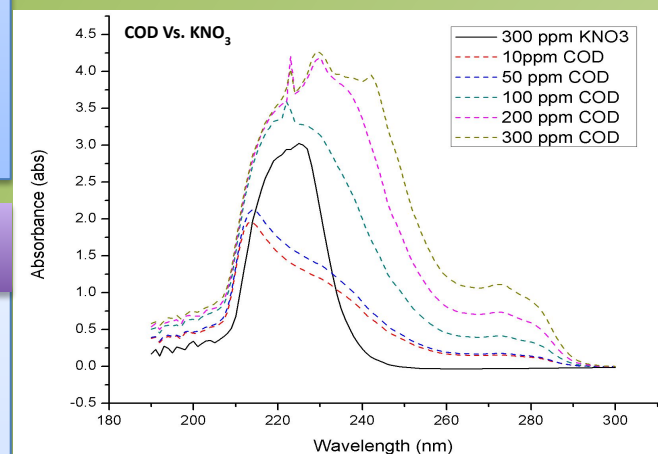
This study would like to acknowledge and appreciate the support of: Dr. Buddhi Rai , Dr. Nathan Withers, Dr. Peter Fisher, Dr. Sally Irwin, Prof. Sean Calder and Chancellor Dr. Lui Hokoana
Bridge to Baccalaureate Program
UH Maui College
B2B program funded by National Science Foundation



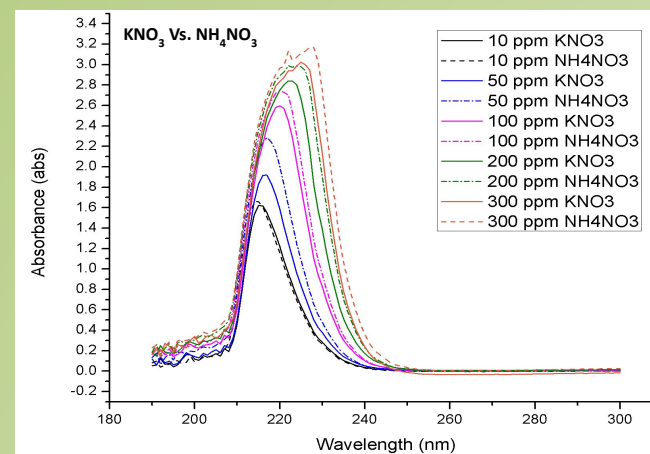
Absorbance vs. wavelength data plot of 10, 50, 100, 200, and 300 ppm concentrations of COD and a baseline of 35 ppt saline solution



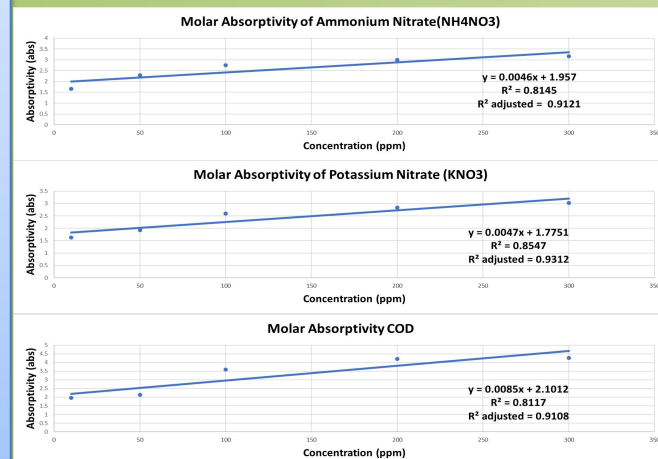
Absorbance vs. wavelength data plot of 10, 50, 100, 200, and 300 ppm concentrations of KNO₃ and a baseline of 35 ppt saline solution



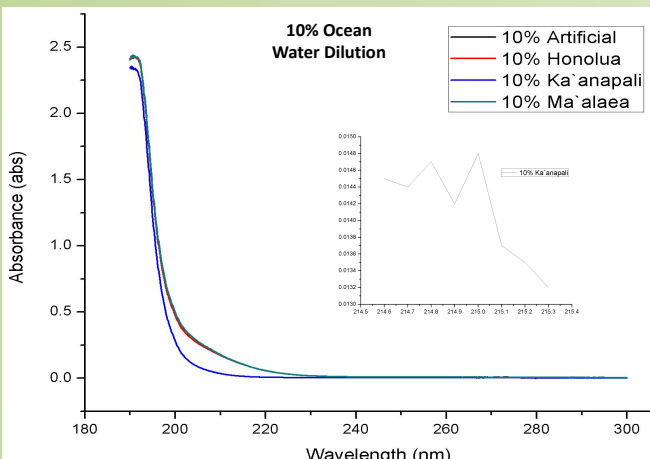
Cross-sensitivity analysis comparing the maximum concentration of KNO₃ with 10, 50, 100, 200, and 300 ppm COD.



Cross-sensitivity analysis comparing the KNO₃ with NH₄NO₃ both at 10, 50, 100, 200, and 300 ppm concentrations.



Comparison of Absorptivity vs. Concentration of KNO₃, NH₄NO₃, and COD to determine the Molar Absorptivity Coefficient (ϵ) of each. These coefficients will then be used to extrapolate concentration of contaminants.



Absorbance vs Wavelength plot of ocean water samples taken from Ka'anapali beach, Ma'alaea harbor, and Honolulu Bay diluted to 10% solution using a deionized water baseline. The inset demonstrates the presence of the peak for Ka'anapali within the sample spectra.

Materials and Procedure

The UV spectra of all solutions and samples were measured in the Genesys spectroscope utilizing a pulse xenon lamp with a slit width of 0.2 nm and quartz cuvettes with a path length of 10 mm. Artificial seawater was prepared using Instant Ocean™ saline mixture and deionized water in concentrations of 1 ppt, 10 ppt, 20 ppt, 30 ppt, and 35 ppt and the samples were compared against deionized water as the spectrophotometric baseline.

For both Nitrate and COD measurements, the baseline of artificial seawater was prepared using the same method at a concentration of 35 ppt. The concentrations of Nitrate solutions to be tested were made from Potassium Nitrate (KNO₃) as well as Ammonium Nitrate (NH₄NO₃) and COD solutions to be tested were made from Potassium Hydrogen Pthalate (KHP) all measured at 10 ppm, 50 ppm, 100 ppm, 200 ppm, and 300 ppm.

Data Analysis and Outcomes

The peak wavelengths ($\lambda_{\max}$) of the KNO₃ and NH₄NO₃ were compared and determined to be indistinguishable from each other in the ultraviolet and visual spectra save for a characteristic peak at the 222 nm wavelength for NH₄NO₃ which was only present at concentrations of 300 ppm or above. Thus, the determination of specific nitrates in the UV/VIS spectrum is not possible except for very high concentration unlikely found in natural systems. However, the range of 215-225 nm could be used to determine the presence of nitrates in unknown solutions.

The peaks of KNO₃ and COD were compared and it was determined that even at very high concentrations of nitrates the spectrum of COD blanketed the entirety of the Nitrate spectrum. This counter indicated previous findings^[2]. Here, however, was a second peak found to be helpful to distinguish COD from Nitrate in the 270-275 nm range.

After realizing the possible interference of external variables the samples and artificial sea water solution were both diluted to 10% and then to 1% and analyzed to find the local maxima of each spectrum. Then, the wavelengths of the local maxima were compared with the selected wavelengths of both Nitrate and COD to determine water quality.

The $\lambda_{\max}$ of the sample spectra were much less pronounced in the diluted samples. A comparison of the slope at different wavelengths of the sample spectra data sets provided a way to determine the various peaks of the ocean water samples. The wavelengths of the local maxima were then compared those of Nitrate and COD. Using the $\lambda_{\max}$ of the sample spectra the concentration was then able to be determined using a transformation of Beer's Law

$$A = \log_{10} \frac{I_0}{I} = \epsilon l c$$

Within the sample spectra peaks of possible contaminants were found, though very indistinct from the rest of the curve, the maxima were within the established wave band of the possible contaminants. This study plans to further investigate these findings and confirm whether or not these indications are in fact pointing to the contaminants being tested for

Looking Forward

The research has shown that in a controlled environment it is possible to simulate indication of nitrates and COD present in ocean water samples. Spectroscopy provides accurate and precise data. However, provided available resources we plan to investigate further using High Pressure Liquid Chromatography (HPLC) as an alternative method of analysis to corroborate of our findings and increase confidence of data from future experimentation. Persistent sampling can rule out environmental factors and establish a baseline from which more accurate data can be extrapolated. Parameters beyond those examined within the scope of this current research will also be considered in order to rule out possible interference leading to shifts in the wavelengths of spectra or obfuscated spectral data